INTERVAL LAB

The 12 distances · practice poster

Everything in music is a distance. On the guitar, distance is frets — 1 fret = 1 semitone. Learn the 12 distances and the whole instrument unlocks.

1 HEAR
Ear tab: name the distance. Hook it to its song.

2 SING
Sing it back first — your voice is the most honest tuner.

3 FIND
Read the neck like a ruler; hunt intervals against the drone.

4 PLAY
Practice games: find every ♭3 from a root, play what you hear.

THE 12 DISTANCES

0	R	unison	1	♭2	minor 2nd	2	2	major 2nd	3	♭3	minor 3rd
4	3	major 3rd	5	4	perfect 4th	6	♭5	tritone	7	5	perfect 5th
8	♭6	minor 6th	9	6	major 6th	10	♭7	minor 7th	11	7	major 7th

12 P8 octave — the same note, one register up · the neck repeats every 12 frets

THE 3 FAMILIES

PERFECT
unison · 4th · 5th · octave — **hollow, strong, stable.** The frame of the building.

IMPERFECT
3rds · 6ths — **the colours.** Major = bright, minor = dark. They make chords happy or sad.

DISSONANT
2nds · 7ths · tritone — **tension that wants to resolve.** Melodies and blues live here.

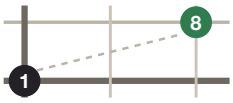
DERIVE — DON'T MEMORISE

1. **Pick 1** — any note, any string. "1" is home.
2. **Count steps** → intervals (3 frets = ♭3).
3. **4 notes spanning a 4th** → a tetrachord.
4. **Lower + gap + upper** → a scale. No table needed.
5. **1st + 3rd note of each half** → the 7th chord inside the scale.
6. **Stack 5ths** → the circle of fifths, keys, everything.

THE GUITAR TRICK · TWO MAGIC SHAPES



Tetrachord: the 4 sits right above 1 — same fret, next string



Octave: the 8 is 2 strings up + 2 frets

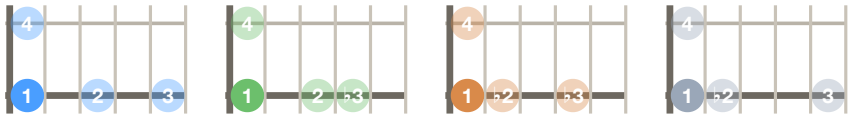
Across the G→B pair add 1 fret — those two strings are a major 3rd apart, not a perfect 4th. Everything else on the neck is a perfect 4th.

CIRCLE OF 5THS · STACK THEM UP

C Am	**G** Em	**D** Bm	**A** F♯m	**E** C♯m	**B** G♯m
F♯ E♭m	**D♭** B♭m	**A♭** Fm	**E♭** Cm	**B♭** Gm	**F** Dm

Top = major key · bottom = its relative minor. Same slice = same notes. Each 5th up adds a ♯, each 5th down adds a ♭ — after F it wraps back to C.

TETRACHORD PATTERNS · 4 notes spanning a perfect 4th



MAJOR W W H **MINOR** W H W **PHRYGIAN** H W W **HARMONIC** H A2 H

Bonus shapes: Lydian W W W (0–2–4–6, spans a tritone) · Diminished H W H (0–1–3–4, spans a major 3rd)

Read it: 1 at fret 0 — walk the W/H steps on the low string, and the **4 sits right above 1** on the next string. Two of these + a whole-step gap = a full scale.

SCALES = LOWER + GAP + UPPER

SCALE	LOWER	GAP	UPPER	CHORD	NOTES
Major	Major	W	Major	Cmaj7	C D E F | G A B C
Natural minor	Minor	W	Phrygian	Am7	A B C D | E F G A
Dorian	Minor	W	Minor	Dm7	D E F G | A B C D
Phrygian	Phrygian	W	Phrygian	Em7	E F G A | B C D E
Mixolydian	Major	W	Minor	G7	G A B C | D E F G
Lydian	Lydian	H	Major	Cmaj7	C D E F♯ | G A B C
Harmonic minor	Minor	W	Harmonic	Am(maj7)	A B C D | E F G♯ A
Diminished	Dim.	W	Dim.	C°7	C D♭ E♭ E | F♯ G A B♭ C

CHORDS · the 7th chord is hiding in the scale — 1st + 3rd note of each half

maj7 1 3 5 7 Major, Lydian	**m7** 1 ♭3 5 ♭7 Minor, Dorian, Phrygian	**7** 1 3 5 ♭7 Mixolydian
m(maj7) 1 ♭3 5 7 Harmonic minor	**°7** 1 ♭3 ♭5 6 Diminished	**triads** 1 3 5 · 1 ♭3 5 · 1 5 major · minor · power

How: split the scale at the gap. Take the 1st + 3rd note of the lower half and the 1st + 3rd of the upper half — those 4 notes are the 7th chord. Every other chord is just a piece of it.

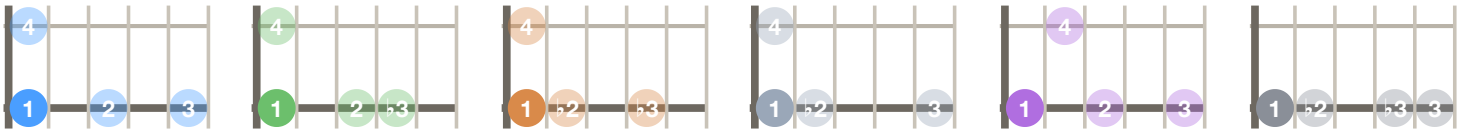
TRY IT NOW: from any note walk W W H W W W H — you just built the major scale.

FRETBOARD SHAPES

tetrachords · scales · chords — for easy identification

Every shape here assumes **perfect-4th string pairs** (E–A, A–D, D–G, B–E). Across the **G→B pair, shift every upper-string note +1 fret**.

TETRACHORDS · THE FOUR LADDERS



MAJOR W W H
0 – 2 – 4 – 5

MINOR W H W
0 – 2 – 3 – 5

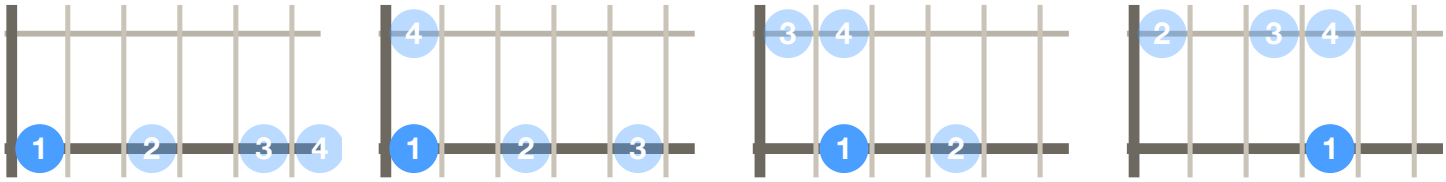
PHRYGIAN H W W
0 – 1 – 3 – 5

HARMONIC H A2 H
0 – 1 – 4 – 5

LYDIAN W W W
0 – 2 – 4 – 6 · spans a tritone

DIMINISHED H W H
0 – 1 – 3 – 4 · spans a major 3rd — all four stay on one string

THE SAME TETRACHORD IN ITS 4 CROSSINGS · MAJOR



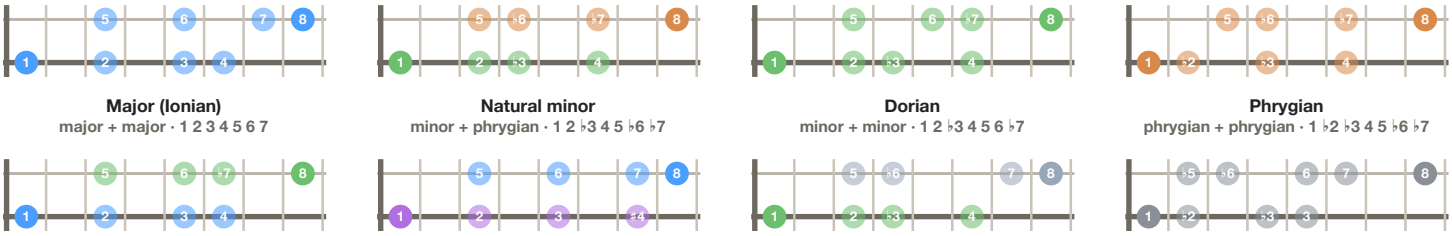
I · all 4 on one string

R · 3 + 1

Z · 2 + 2

J · 1 + 3

SCALES · ONE OCTAVE = TWO TETRACHORDS ON TWO STRINGS



Major (Ionian)
major + major · 1 2 3 4 5 6 7

Natural minor
minor + phrygian · 1 2 ♭3 4 5 ♭6 ♭7

Dorian
minor + minor · 1 2 ♭3 4 5 6 ♭7

Phrygian
phrygian + phrygian · 1 ♭2 ♭3 4 5 ♭6 ♭7

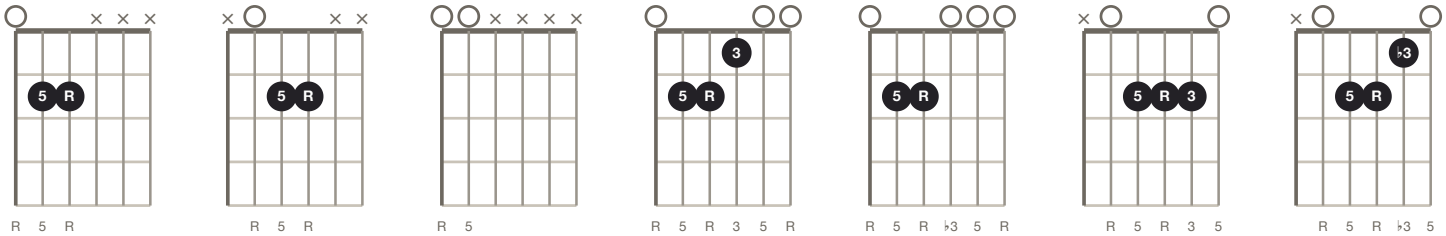
Mixolydian
major + minor · 1 2 3 4 5 6 ♭7

Lydian
lydian + major · 1 2 3 ♯4 5 6 7

Harmonic minor
minor + harmonic · 1 2 ♭3 4 5 ♭6 7

Diminished (whole-half)
dim + dim · 1 ♭2 ♭3 3 ♭5 ♭6 6 7

CHORDS · THE SHAPES AND WHAT THEY'RE MADE OF



E5 R 5 R
power · E-shape

A5 R 5 R
power · A-shape

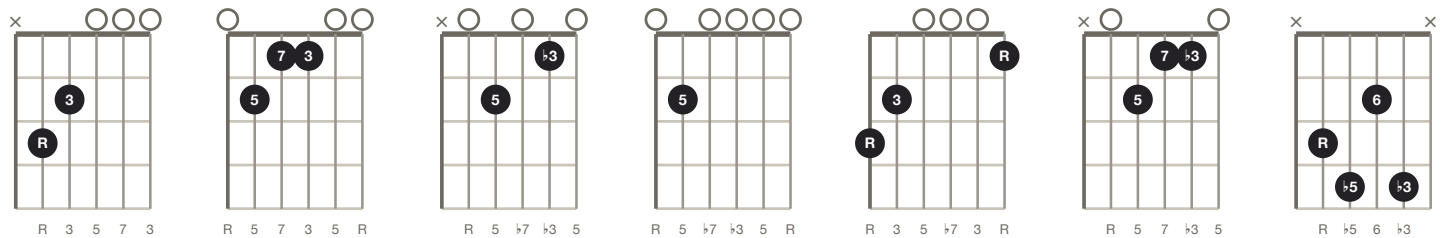
Drop power R 5
one finger · drop tunings

E major R 5 R 3 5 R
E-shape · movable

E minor R 5 R ♭3 5 R
E-shape · movable

A major R 5 R 3 5
A-shape · movable

A minor R 5 R ♭3 5
A-shape · movable



Cmaj7 1 3 5 7
open

Emaj7 1 3 5 7
movable

Am7 1 ♭3 5 ♭7
open

Em7 1 ♭3 5 ♭7
movable

G7 1 3 5 ♭7
open

Am(maj7) 1 ♭3 5 7
harmonic minor's chord

C°7 1 ♭3 ♭5 6
diminished

How to read: filled dot = fretted note (label = its role: R root, 3rd, 5th, ♭7…) · ○ = open string · × = muted · degree row under each diagram = what that string plays · movable shapes are shown at the nut — slide the whole shape anywhere.

OCTAVE TWINS — the same note is always 2 strings up + 2 frets away (3 frets across G→B). Find one note and you've found them all.

SCALES = 2 LADDERS + A GAP — spot the two tetrachords on the neck and you own the whole scale. Chords are just the 1st + 3rd note of each ladder.

IDENTIFY, DON'T MEMORISE: `find the 1` → `walk the steps` → `the shape names itself.`

FRETBOARD MAPS

standard E · drop C — every note, every interval

How to use: pick any fret as **1** → the top ruler names every distance from it. The small label under each note = that note's distance from the **open string**. Green = octave (12 frets = same note). The neck repeats every 12 frets.

STANDARD E · E2 · A2 · D3 · G3 · B3 · E4

strings low → high · G → B = major 3rd (every other pair is a perfect 4th)

RULER	R	♭2	2	♭3	3	4	♭5	5	♭6	6	♭7	7	P8
FRET	0	1	2	3·	4	5·	6	7·	8	9·	10	11	12··
E4	E 1	F ♭2	F# 2	G ♭3	G# 3	A 4	A# ♭5	B 5	C ♭6	C# 6	D ♭7	D# 7	E 8
P4													
B3	B 1	C ♭2	C# 2	D ♭3	D# 3	E 4	F ♭5	F# 5	G ♭6	G# 6	A ♭7	A# 7	B 8
M3													
G3	G 1	G# ♭2	A 2	A# ♭3	B 3	C 4	C# ♭5	D 5	D# ♭6	E 6	F ♭7	F# 7	G 8
P4													
D3	D 1	D# ♭2	E 2	F ♭3	F# 3	G 4	G# ♭5	A 5	A# ♭6	B 6	C ♭7	C# 7	D 8
P4													
A2	A 1	A# ♭2	B 2	C ♭3	C# 3	D 4	D# ♭5	E 5	F ♭6	F# 6	G ♭7	G# 7	A 8
P4													
E2	E 1	F ♭2	F# 2	G ♭3	G# 3	A 4	A# ♭5	B 5	C ♭6	C# 6	D ♭7	D# 7	E 8

DROP C · C2 · G2 · C3 · F3 · A3 · D4

strings low → high · F → A = major 3rd (every other pair is a perfect 4th)

RULER	R	♭2	2	♭3	3	4	♭5	5	♭6	6	♭7	7	P8
FRET	0	1	2	3·	4	5·	6	7·	8	9·	10	11	12··
D4	D 1	D# ♭2	E 2	F ♭3	F# 3	G 4	G# ♭5	A 5	A# ♭6	B 6	C ♭7	C# 7	D 8
P4													
A3	A 1	A# ♭2	B 2	C ♭3	C# 3	D 4	D# ♭5	E 5	F ♭6	F# 6	G ♭7	G# 7	A 8
M3													
F3	F 1	F# ♭2	G 2	G# ♭3	A 3	A# 4	B ♭5	C 5	C# ♭6	D 6	D# ♭7	E 7	F 8
P4													
C3	C 1	C# ♭2	D 2	D# ♭3	E 3	F 4	F# ♭5	G 5	G# ♭6	A 6	A# ♭7	B 7	C 8
P4													
G2	G 1	G# ♭2	A 2	A# ♭3	B 3	C 4	C# ♭5	D 5	D# ♭6	E 6	F ♭7	F# 7	G 8
P4													
C2	C 1	C# ♭2	D 2	D# ♭3	E 3	F 4	F# ♭5	G 5	G# ♭6	A 6	A# ♭7	B 7	C 8

THE RULER — the top row is every distance from any fret you pick as 1. Slide it anywhere; the labels never change. 1 fret = 1 semitone.

OCTAVE TWINS — the same note sits 2 strings up + 2 frets away. Cross the major-3rd pair (G→B · F→A) and add +1 fret. Find one copy, find them all.

FRET-5 RULE — the next string's open note = this string at fret 5 — fret 4 across the major-3rd pair. That single rule builds both tunings.

FRET 12 = OCTAVE — every string repeats after 12 frets, so this board is the whole neck, forever. Note names: white = natural · amber = ♯/♭ · C♯ = D♭.

DRILL: `name a note` → `find every copy` → `check with octave twins`